Celo

Internet Face Assessment

Prepared by: **Halborn**

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Visit: **Halborn.com**

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DOCUMENT REVISION HISTORY

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EXECUTIVE OVERVIEW

1.1 INTRODUCTION

cLabs engaged Halborn to conduct an Internet Face Assessment beginning on February 9th, 2022 and ending on March 22nd, 2022 . The security assessment was scoped to all the Celo network nodes.

1.2 ASSESMENT SUMMARY

The team at Halborn was provided six weeks for the engagement and assigned a full-time security engineer to audit the security of the smart contract. The security engineer is a blockchain and smart-contract security expert with advanced penetration testing, smart-contract hacking, and deep knowledge of multiple blockchain protocols.

The purpose of this engagement is:

- Confirm the privacy of the validators IP address
- Identify possible misconfigured nodes
- Identify cloud provider distribution
- Uncover any potential risk of centralization

For this purpose, Halborn deployed a validator on the baklava network and on the mainnet. Analyzed networking connections. Scan the IP addresses connected to the machines. And finally perform the analysis from the data obtained detailed in this report.

1.3 TEST APPROACH & METHODOLOGY

To accomplish this task, Halborn set up a Proxy and a Validator and examined the network connections made by the machines to study the communication protocol and visibility between the different nodes on the network.

- Configure Account node, Validator node and Proxy node on baklava network and on mainnet
- Carry out traffic analysis in each of the different configured nodes
- Extract all the IP addresses connected to each of the different machines
- Analyze all IP addresses focusing on possible misconfigurations on the exposed surface

The set of tools used during this assessment are:

- AWS cloud provider for validator deployment
- Scapy, tcpdump and wireshark for traffic analysis
- Nmap for the exposed surface of the nodes
- Celo Network official documentation and other resources

1.4 SCOPE

The scope of this assessment is any machine involved in the Celo Network. Focusing on validators and full nodes.

FINDINGS & TECH DETAILS

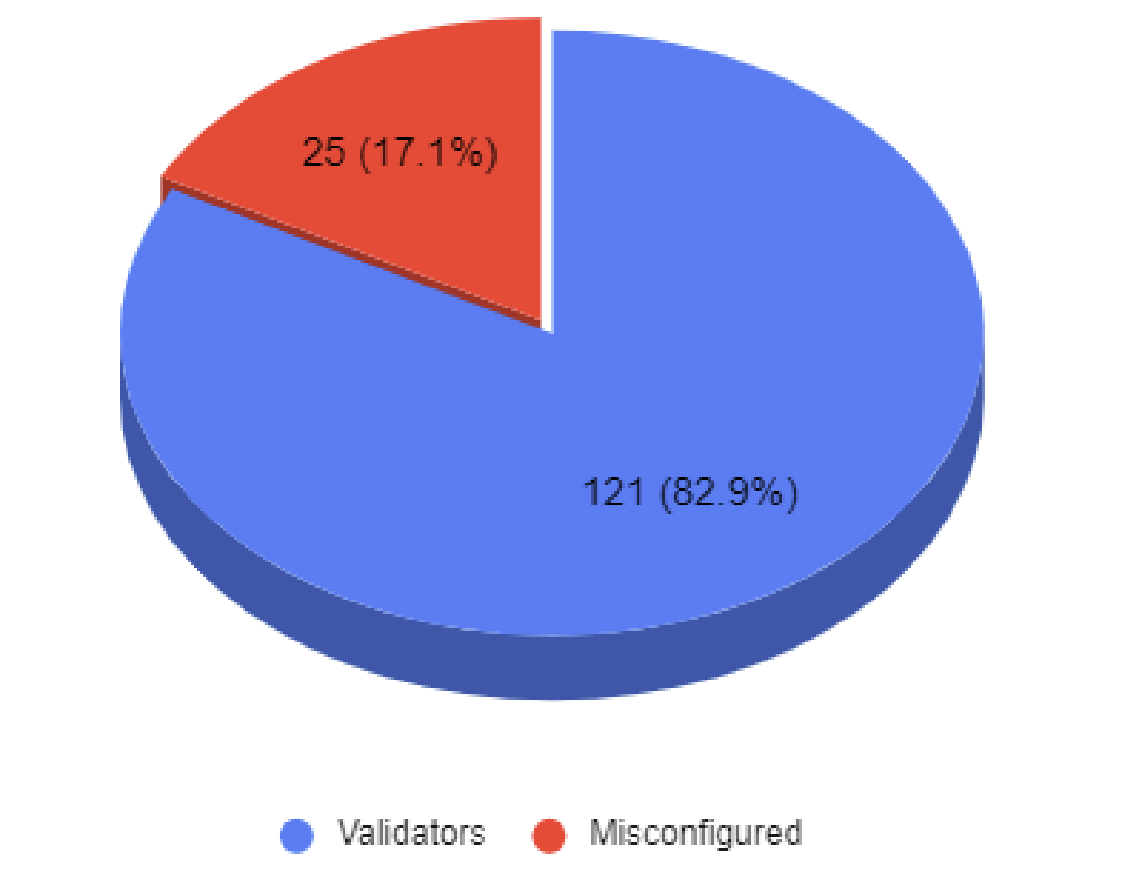
2.1 NODES OVERVIEW

Description:

The validators have an architecture of two different machines. These are the proxy and the validator itself. The proxy is responsible for handling communication with the network and forwarding the appropriate messages to the validator. This architecture prevents the validator from being exposed to the public internet and prevents DoS and other potential attacks that can compromise funds locked in it.

Results:

As can be seen in the following graph, **the mainnet network has 25 validators that do not follow the standard operations.**



This data is obtained from the total number of validators counted on the stats celo web page, a total of 146, and the proxies exposing port 30503 found during this assessment, it is about 25 proxies. This metric assumes that any of the obtained IP addresses containing port 30503 is a proxy.

Nearly one in five nodes has a non-compliant setup. This configuration poses a risk due to misconfiguration, which opens the attack surface to external threat actors.

Finally, it is important to consider that of the 146 validators in the network, there are 110 chosen validators. This study does not filter between elected and non-elected validators.

2.2 CLOUD PROVIDER DISTRIBUTION

Description:

Knowing the distribution summary of cloud providers is interesting for several reasons. First, cLabs can get an overview of the infrastructures the network is based on. The cLabs team can design a recovery plan in case of failure of them if required.

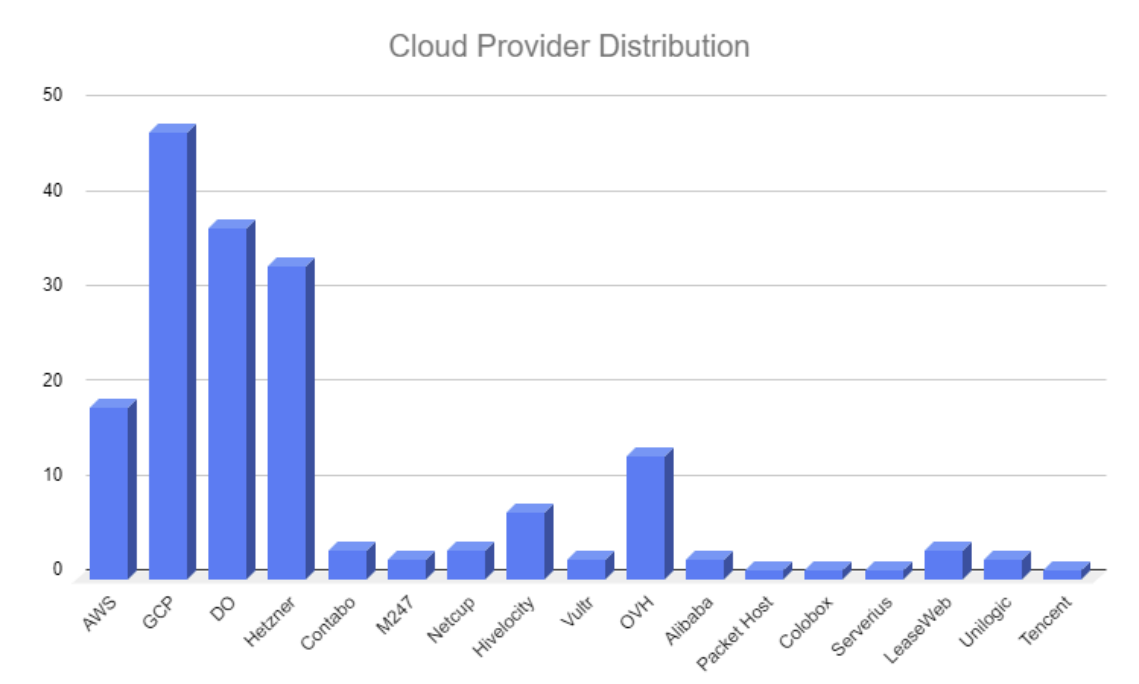
Results:

14 different cloud providers have been detected on the identified machines running on the mainnet.

- 18 IP addresses on AWS
- 37 IP addresses on Digital Ocean
- 47 IP addresses on Google Cloud
- 33 IP addresses on Hetzner
- 1 IP address on Azure
- 3 IP addresses on Contabo
- 2 IP addresses on M247
- 3 IP addresses on Netcup
- 7 IP addresses on Hivelocity
- 2 IP addresses on Vultr
- 13 IP addresses on OVH
- 2 IP addresses on Alibaba
- 1 IP address on Packet Host
- 1 IP address on Colobox
- 1 IP address on Serverius
- 3 IP addresses on LeaseWeb
- 2 IP addresses on Unilogic
- 1 IP address on Tencent
- 1 IP address on Wired ISP
- 1 IP address on Opera Software
- 1 IP address Vanilla Telcoms
- 1 IP address on Online

This information shows that Celo Network has a diversified cloud distribution. However, the most used cloud providers are Google Cloud, Digital Ocean and Hetzner. It is important to remark that this distribution corresponds to both Validators and Full Nodes.

The following graph shows the distribution of the nodes in the different cloud providers.



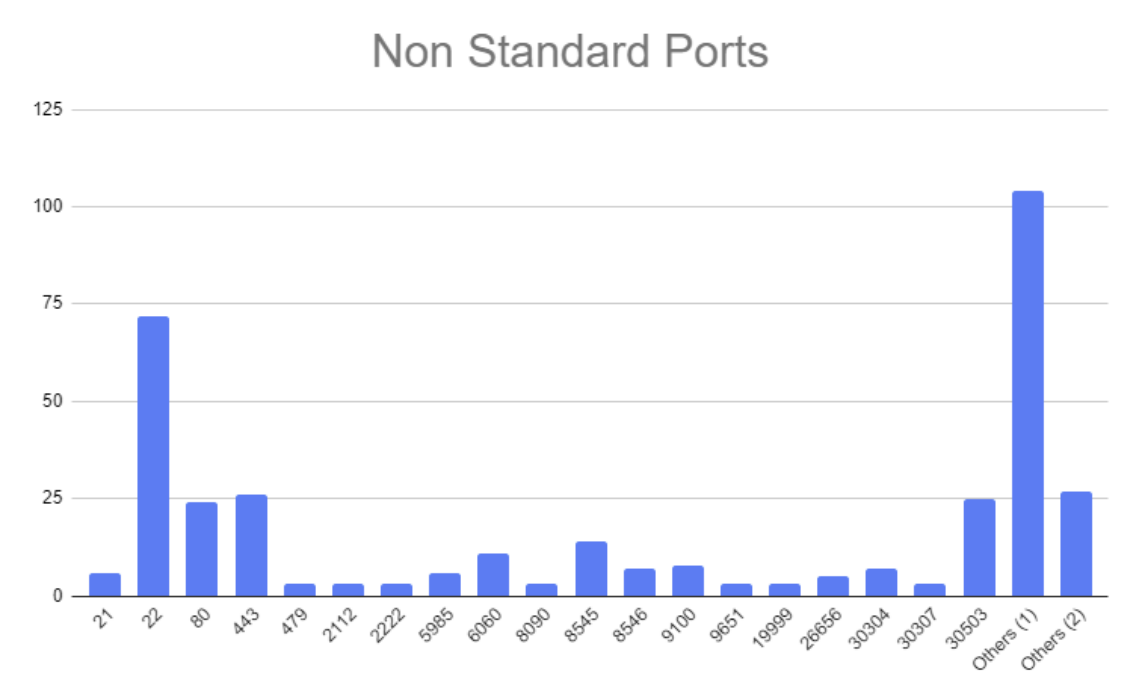
Recommendation:

While there is a strong position on trusted cloud providers, it could be a good suggestion to provide a list of trusted and untrusted providers based on the cLabs knowledge of them. This means avoiding the use of untrusted third-party cloud providers.

2.3 PORT DISTRIBUTION OVERVIEW

Description:

All 181 IP addresses used in this study contain open port 30303. However, to clearly expose the surface of non-standard nodes, the following graph is included in the report.



The graph includes two bars that do not correspond to any specific port number. Others(1) correspond to different unique ports that were matched only once. While Others(2) correspond to unique different ports that appeared twice, the reason behind this is that these unique ports made a graphic not understandable.

Results:

Knowing the internal configuration of the Celo Node according to the official documentation, the exposed ports that may be more risky are:

- 8545 RPC-HTTP: 14 exposed ports

- 8546 WS-RPC: 7 exposed ports
- 6060 HTTP Metrics: 11 exposed ports

These ports should be accessible from a private network or from localhost. For port 8545, quoting the official documentation: **Do not expose this port to the public internet!**. For port 6060: **Make sure you never expose the pprof service to the public internet.**

It is important to remark that no further research has been done to validate if the open service was really what it was supposed to be. According to cLabs guidelines, there has been no interaction for detection besides a TCP handshake and basic detection with the exposed services.

2.4 VALIDATORS DISCOVERY APPROACH

Description:

To discover the hidden validators behind the Celo Network proxies, two main approaches has been considered.

- Study the communication protocol of the network searching for possible leaks
- Given the search for misconfigured proxies for the associated misconfigured validators

Results:

While a light manual code review of the nodes' source code repository has been tested, no data structure containing the IP addresses of validators on the network has been identified. So to search for the actual validators, the second approach is more viable.

Knowing the proxies with port 30503 open, it is possible to infer that the validators will also be exposed. Also, knowing the range of the cloud provider, it is feasible to think that the validator is in the same range as the proxy.

So, taking the IP address 35.212.191.229 of the baklava network as an example, that:
- Contains port 30303 and 30503 open
- Placed on GCP us-west1

GCP us-west-1 contains 21 IP ranges as specified here. The range scan revealed a list of 32 IP addresses that contained port 30303 open. From these 32, it is possible to filter 4 IP addresses that appear in the mainnet scan and also remove the 3 known proxies that are from the baklava network. This reduces the number to 25 possible IP addresses.

After scanning all open ports for all 25 IP addresses, it was not possible to narrow down the list to determine the actual validator. This list of

25 IP addresses can be found in Appendix A. However, there may be two possible ways to do this:

- Attempt more aggressive scans to open services
- Reverse the communication protocol to find a specific packet to which the validator responds

This same approach can be done on any cloud provider. Therefore, finding the IP address of a misconfigured validator is feasible under certain circumstances.

2.5 TRAFFIC ANALYSIS

Description:

To validate the actual communication of each machine in a validator, the traffic was monitored, dumped and studied.

From the proxy analysis, all the different IP addresses of the other validators and full nodes of the network were obtained. To accomplish this goal, the following script was developed after dumping the machine's traffic with tcpdump on the corresponding interface.

Listing 1: extractips.py

```
from scapy.all import *

pkts=rdpcap(sys.argv[1])

ips=[]
for pkt in pkts
    if pkt[IP].src not in ips:
        ips.append(pkt[IP].src)
    if pkt[IP].dst not in ips:
        ips.append(pkt[IP].dst)

ips = list(set(ips))
for ip in ips:
    print(ip)
```

Results:

The two main traffic dumps that revealed important information for this assessment are:

Name	MD5 (Hash)
baklava-validator.raw	1102026ccb83e8770a6a4dd51d577d0a
main-proxy.raw	39b6be5e1ec494b298777d5b97d53432

The main-proxy.raw dump provided all the IP addresses used for the mainnet statistics analysis of this report. This includes: cloud distribution, node overview and node footprint.

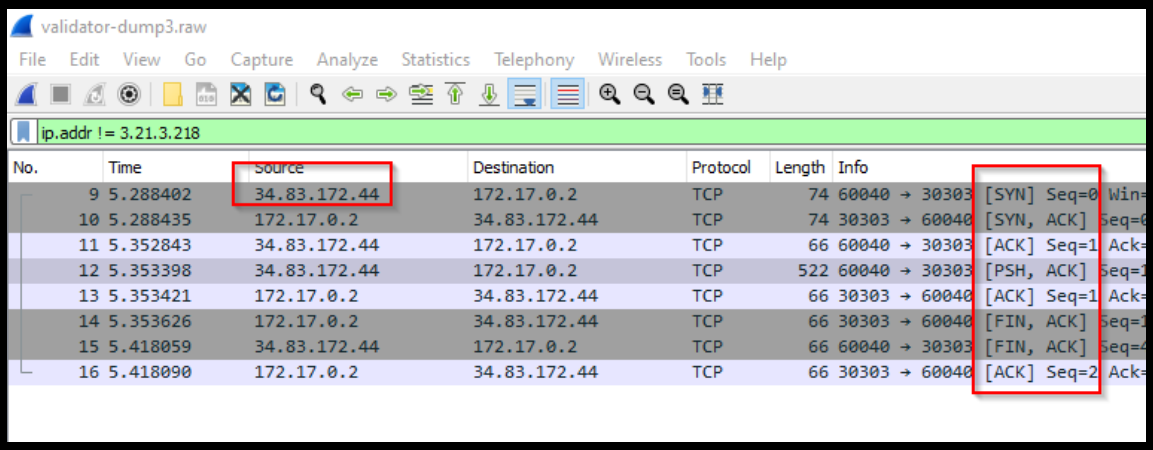
From baklava-validator.raw an unexpected validator network communication was detected, leading to an information disclosure finding on an unidentified network component.

This same procedure was repeated in mainnet. The IP's obtained from the proxy are the ones used for the statistics shown above. However, the validator did not reveal any communication to the IP addresses revealed in the next section.

EXPOSED UNKNOWN NETWORK COMPONENT:

Description:

One of the main findings of the traffic analysis is the communication of the baklava validator with some other machine than the proxy.



This is an unexpected behavior, as the validator should only have communicated with the proxy. However, it is important to mention that this behavior was not reproduced in the mainnet validator.

The IP address exposed in the image above was scanned and revealed some data that could point to a telemetry system. However, this information has not yet been validated. This machine exists on the baklava network and the mainnet with different IP addresses but running the same services

and with the same information exposed.

Results:

Both machines communicate with the following API which also reveals some routes:
- https://104.198.96.218/api/v1/

Although what exactly these machines and APIs are not been confirmed, this seems a central point where information about the state of the network is processed.

Baklava network In the Baklava network, this IP address exposed in the traffic analysis is:
- 34.83.172.44

The IP addresses contained port 10250 with HTTPS and port 10255 with HTTP. Some protected directories have been found on the HTTPS port that were not properly protected and were revealing private information about docker containers within the HTTP service. These directories are:

- http://34.83.172.44:10255/pods
- http://34.83.172.44:10255/metrics

Mainnet network To find the peer machine on mainnet, all IP addresses obtained from the dump were scanned for open ports 10250 and 10255. The following IP address revealed that surface:
- 34.105.47.65

The IP address also contained the same misconfiguration regarding the HTTP service and sensitive directories:

- http://34.105.47.65:10255/pods
- http://34.105.47.65:10255/metrics

These resources have been succesfully protected following Halborn's notification to cLabs.

2.6 Non-Standard Nodes

Port	Service	Version
8545	unknown	-
30303	tcpwrapped	-

107.20.177.242

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.14.0 (Ubuntu)
8302	unknown	-
30303	tcpwrapped	-
39797	-	-

209.126.9.140

Port	Service	Version
30303	tcpwrapped	-
30503	tcpwrapped	-

116.202.157.121

Port	Service	Version
21	ftp	Microsoft ftpd
80	http	Microsoft IIS httpd 10.0

5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-

46.21.149.114

Port	Service	Version
6060	http	Golang net
9100	jetdirect?	-
30303	tcpwrapped	-
30503	tcpwrapped	-
40511	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)

78.141.213.188

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

134.209.32.34

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.18.0 (Ubuntu)
443	ssl	http

10050	tcpwrapped	-
19999	dnp-sec?	-
30303	tcpwrapped	-

135.181.116.108

Port	Service	Version
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
30303	tcpwrapped	-

138.197.137.57

Port	Service	Version
2112	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

135.181.96.236

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
1789	hello?	-
30303	tcpwrapped	-
30503	tcpwrapped	-

47.74.17.221

Port	Service	Version
80	http	nginx
443	ssl	http
30303	tcpwrapped	-

49.12.1.140

Port	Service	Version
443	ssl	http
30303	tcpwrapped	-

212.32.242.72

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

35.231.227.29

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.18.0 (Ubuntu)
443	ssl	http

9090	ssl	zeus-admin?
30303	tcpwrapped	-

161.97.131.43

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
6060	http	Golang net
9001	ssl	tor-orport?
9100	jetdirect?	-
30303	tcpwrapped	-
30503	tcpwrapped	-

66.165.233.202

Port	Service	Version
80	http	nginx
443	ssl	http
2222	ssh	OpenSSH 8.2p1 (protocol 2.0)
30303	tcpwrapped	-

194.13.81.9

Port	Service	Version
80	http	nginx 1.14.0 (Ubuntu)

443	ssl	http
2222	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

178.63.23.24

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
30303	tcpwrapped	-

139.178.82.109

Port	Service	Version
80	http	nginx 1.14.0 (Ubuntu)
443	ssl	http
3003	http	Node.js Express framework
6379	redis	Redis key-value store
8140	ssl	http
8545	daap	mt-daapd DAAP
30303	tcpwrapped	-
64592	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)

47.52.106.149

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

35.229.87.81

Port	Service	Version
30303	tcpwrapped	-
30304	tcpwrapped	-

138.201.81.132

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.18.0 (Ubuntu)
8545	http	Golang net
30303	tcpwrapped	-

142.132.132.249

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
27656	-	-

30303	tcpwrapped	-
30503	tcpwrapped	-
30656	-	-
36656	-	-

51.91.67.48

Port	Service	Version
22	ssh	OpenSSH 8.4p1 Ubuntu 5ubuntu1.2 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

18.220.93.105

Port	Service	Version
6060	http	Golang net
17549	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

54.38.177.35

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
3030	arepa-cas?	-

19999	dnp-sec?	-
24567	-	-
30303	tcpwrapped	-

195.201.204.239

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
8545	http	Golang net
30303	tcpwrapped	-

143.198.70.46

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
6060	http	Golang net
30303	tcpwrapped	-
30503	tcpwrapped	-

54.36.123.27

Port	Service	Version
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
30303	tcpwrapped	-

128.199.116.107

Port	Service	Version
22	ssh	OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
80	http	Caddy httpd
443	ssl	https
4369	epmd	Erlang Port Mapper Daemon
7878	http	Node.js Express framework
8080	http	Node.js Express framework
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
8549	unknown	-
8550	unknown	-
9545	http	Golang net
9546	daap	mt-daapd DAAP
9650	unknown	-
9651	-	-
30303	tcpwrapped	-
30304	tcpwrapped	-

141.95.35.173

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

34.82.177.159

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

35.230.51.149

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
6060	http	Golang net
8545	daap	mt-daapd DAAP
30303	tcpwrapped	-

65.108.93.26

Port	Service	Version
80	http	nginx 1.18.0 (Ubuntu)
2857	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

178.128.228.96

Port	Service	Version

22	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
30303	tcpwrapped	-

34.122.195.160

Port	Service	Version
8008	http	-
30303	-	-

89.149.218.141

Port	Service	Version
5432	postgresql	PostgreSQL DB 9.6.0 or later
27000	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

5.9.1.137

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

34.75.171.119

Port	Service	Version
22	ssh	OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
30303	tcpwrapped	-

35.198.155.25

Port	Service	Version
22	ssh	OpenSSH 8.3p1 Ubuntu 1ubuntu0.1 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

37.120.245.147

Port	Service	Version
80	http	Node.js Express framework
30303	tcpwrapped	-
30304	tcpwrapped	-

95.217.74.75

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
10090	http	nginx 1.18.0 (Ubuntu)
30303	tcpwrapped	-

54.196.8.154

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

185.161.189.110

Port	Service	Version
80	http	Apache httpd 2.4.29 ((Ubuntu))
443	ssl	http
30303	tcpwrapped	-

52.33.96.198

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

44.225.95.57

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
2200	ssl	http
2204	ssl	http

2210	ssl	http
2214	ssl	http
2220	ssl	http
8010	nagios-nsca	Nagios NSCA
9210	oma-mlp?	-
25000	ssl	http
30303	tcpwrapped	-
30305	tcpwrapped	-
30306	tcpwrapped	-
30307	tcpwrapped	-

82.145.220.247

Port	Service	Version
479	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

143.198.36.58

Port	Service	Version
30302	tcpwrapped	-
30303	tcpwrapped	-
31312	libp2p-multistream	libp2p multistream protocol 1.0.0
32346	-	-

3.129.250.135

Port	Service	Version
2112	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
9100	jetdirect?	-
30303	tcpwrapped	-
46060	http	Golang net

95.216.206.183

Port	Service	Version
30303	tcpwrapped	-
30503	tcpwrapped	-
30656	-	-

54.38.46.124

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

51.89.194.154

Port	Service	Version

22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

34.82.242.97

Port	Service	Version
22	ssh	OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
9650	unknown	-
9651	-	-
30303	tcpwrapped	-

136.243.40.207

Port	Service	Version
2222	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

135.181.18.216

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

35.234.9.93

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

95.217.134.199

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
80	http	nginx
443	ssl	http
21834	http	Golang net
24966	http	Golang net
26445	http	Golang net
26656	unknown	-
30002	pago-services2?	-
30003	amicon-fpsu-ra?	-
30008	-	-
30009	-	-
30010	unknown	-
30011	tcpwrapped	-
30016	tcpwrapped	-
30018	tcpwrapped	-
30020	tcpwrapped	-

30021	unknown	-
30022	tcpwrapped	-
30026	tcpwrapped	-
30027	tcpwrapped	-
30028	http	Golang net
30029	http	Golang net
30037	-	-
30038	-	-
30303	tcpwrapped	-
30307	tcpwrapped	-

135.125.160.122

Port	Service	Version
30303	tcpwrapped	-
30503	tcpwrapped	-

185.181.103.138

Port	Service	Version
6060	http	Golang net
9100	jetdirect?	-
15609	ssh	OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

49.12.47.105

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
30303	tcpwrapped	-

34.135.14.9

Port	Service	Version
21	ftp	Microsoft ftpd
80	http	Microsoft IIS httpd 10.0
5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-

46.21.149.74

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

34.77.108.219

Port	Service	Version

22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

13.212.225.86

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.6 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

35.225.183.134

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

35.197.109.39

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

34.83.16.7

Port	Service	Version

21	ftp	Microsoft ftpd
80	http	Microsoft IIS httpd 10.0
135	msrpc	Microsoft Windows RPC
2179	vmrdp?	-
5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-
30305	tcpwrapped	-

23.111.170.170

Port	Service	Version
25	smtp	Postfix smtpd
80	ssl	http
443	ssl	http
4444	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
8082	http	Golang net
9993	palace-2?	-
29994	-	-
30303	tcpwrapped	-
30503	tcpwrapped	-
48454	-	-

51.15.234.235

Port	Service	Version

22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

34.91.183.212

Port	Service	Version
22	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
1317	vrts-ipcserver?	-
5672	amqp	RabbitMQ 3.9.11 (0-9)
8090	http	Jetty 9.4.11.v20180605
8091	http	Jetty 9.4.11.v20180605
8092	http	Jetty 9.4.11.v20180605
8301	amberon?	-
8332	bitcoin-jsonrpc	Bitcoin or Litecoin JSON-RPC
8333	bitcoin?	-
8535	daap	mt-daapd DAAP
8536	daap	mt-daapd DAAP
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
8625	http	Golang net
8626	daap	mt-daapd DAAP
18888	apc-necmp?	-
26656	unknown	-
26657	unknown	-
30303	tcpwrapped	-

30311	tcpwrapped	-
50051	unknown	-
50061	unknown	-
50071	unknown	-

95.217.194.183

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

168.119.141.61

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
443	ssl	http
8090	ssl	http
30303	tcpwrapped	-
30503	tcpwrapped	-

34.74.71.105

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)

443	ssl	http
30303	tcpwrapped	-

35.237.72.81

Port	Service	Version
6060	http	Golang net
9100	jetdirect?	-
30303	tcpwrapped	-
30503	tcpwrapped	-
59681	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)

51.89.42.183

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

51.83.220.86

Port	Service	Version
22	ssh	OpenSSH 7.4p1 Debian 10+deb9u7 (protocol 2.0)
30303	tcpwrapped	-

35.224.62.220

Port	Service	Version
24	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
10003	http	Golang net
30303	tcpwrapped	-
30503	tcpwrapped	-

94.16.110.231

Port	Service	Version
6060	http	Golang net
9100	jetdirect?	-
10307	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

136.244.112.96

Port	Service	Version
22	ssh	OpenSSH 8.4 (protocol 2.0)
26656	unknown	-
26666	unknown	-
30303	tcpwrapped	-
30304	-	-

150.109.148.233

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

35.234.49.253

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.6 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

54.71.153.100

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.14.0 (Ubuntu)
443	ssl	http
30303	tcpwrapped	-

3.12.127.105

Port	Service	Version

2112	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
9100	jetdirect?	-
30303	tcpwrapped	-
46060	http	Golang net

54.36.109.175

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
13337	http	Node.js Express framework
19090	unknown	-
19091	unknown	-
26656	unknown	-
26657	unknown	-
30303	tcpwrapped	-
30304	tcpwrapped	-
32668	tcpwrapped	-
42220	tcpwrapped	-

157.90.92.115

Port	Service	Version
22	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
30303	tcpwrapped	-

34.133.11.111

Port	Service	Version
80	http	Apache httpd 2.4.38 ((Debian))
443	ssl	http
30303	tcpwrapped	-
39418	ssh	(protocol 2.0)

5.189.169.32

Port	Service	Version
21	ftp	Microsoft ftpd
5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-

23.92.70.58

Port	Service	Version
30303	tcpwrapped	-
30503	tcpwrapped	-

95.216.227.246

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)
443	ssl	http

30303	tcpwrapped	-

35.243.178.68

Port	Service	Version
22	ssh	OpenSSH 8.4p1 Ubuntu 5ubuntu1.1 (Ubuntu Linux; protocol 2.0)
6783	http	nginx 1.21.6
19999	dnp-sec?	-
30303	tcpwrapped	-
30333	-	-
30334	-	-

141.94.139.101

Port	Service	Version
479	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

167.99.186.191

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

40.123.252.43

Port	Service	Version
30303	tcpwrapped	-
30313	tcpwrapped	-
30333	tcpwrapped	-

65.21.173.171

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

35.194.243.189

Port	Service	Version
22	ssh	OpenSSH 7.9p1 Debian 10+deb10u2 (protocol 2.0)
8070	ucs-isc?	-
30303	tcpwrapped	-

65.21.76.106

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)

30303	tcpwrapped	-

3.83.232.200

Port	Service	Version
80	http	nginx 1.20.1
9651	-	-
9653	-	-
26660	unknown	-
26663	unknown	-
30303	tcpwrapped	-
30304	tcpwrapped	-
30309	tcpwrapped	-
30310	tcpwrapped	-
30311	tcpwrapped	-
30312	tcpwrapped	-
30315	-	-
30316	-	-
30317	tcpwrapped	-
30318	tcpwrapped	-
30319	-	-
30320	-	-
30321	tcpwrapped	-
30322	tcpwrapped	-
30323	tcpwrapped	-
30324	tcpwrapped	-

30325	tcpwrapped	-

209.195.10.164

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
80	http	nginx 1.18.0 (Ubuntu)
443	ssl	http
10050	tcpwrapped	-
30303	tcpwrapped	-

65.21.226.187

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

51.89.173.205

Port	Service	Version
21	ftp	Microsoft ftpd
5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-

23.111.187.74

Port	Service	Version
9000	cslistener?	-
30303	tcpwrapped	-
30304	tcpwrapped	-
30306	-	-

95.217.203.43

Port	Service	Version
6060	http	Golang net
9100	jetdirect?	-
13932	ssh	OpenSSH 8.4p1 Debian 5 (protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

95.217.224.234

Port	Service	Version
479	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

167.99.187.175

Port	Service	Version

26	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.6 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

37.120.187.56

Port	Service	Version
5665	unknown	-
30303	tcpwrapped	-
30503	tcpwrapped	-
62543	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)

167.172.188.4

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.4 (Ubuntu Linux; protocol 2.0)
6060	http	Golang net
8545	daap	mt-daapd DAAP
30303	tcpwrapped	-

78.47.83.176

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.2 (Ubuntu Linux; protocol 2.0)

80	http	nginx
443	ssl	http
21171	http	Golang net
26656	unknown	-
29227	http	Golang net
30002	pago-services2?	-
30003	amicon-fpsu-ra?	-
30008	-	-
30009	-	-
30010	unknown	-
30011	tcpwrapped	-
30016	tcpwrapped	-
30018	tcpwrapped	-
30020	tcpwrapped	-
30021	unknown	-
30022	tcpwrapped	-
30026	tcpwrapped	-
30027	tcpwrapped	-
30028	http	Golang net
30029	http	Golang net
30037	-	-
30038	-	-
30303	tcpwrapped	-
30307	tcpwrapped	-
31623	http	Golang net

135.125.160.121

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

34.86.41.108

Port	Service	Version
8545	daap	mt-daapd DAAP
8546	daap	mt-daapd DAAP
30303	tcpwrapped	-

138.197.133.143

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-
30503	tcpwrapped	-

34.94.124.30

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)

80	http	nginx 1.18.0 (Ubuntu)
30303	tcpwrapped	-

135.181.178.182

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
6060	http	Golang net
8545	daap	mt-daapd DAAP
30303	tcpwrapped	-

65.108.88.31

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

34.90.111.77

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

15.165.59.122

Port	Service	Version
27000	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

213.239.195.206

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
443	ssl	http
30303	tcpwrapped	-

35.194.200.224

Port	Service	Version
21	ftp	Microsoft ftpd
80	http	Microsoft IIS httpd 10.0
5985	http	Microsoft HTTPAPI httpd 2.0 (SSDP
30303	tcpwrapped	-

23.111.189.2

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.3 (Ubuntu Linux; protocol 2.0)
443	ssl	http

30303	tcpwrapped	-

35.193.145.229

Port	Service	Version
22	ssh	OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
443	ssl	http
8090	ssl	http
30303	tcpwrapped	-

95.216.100.231

Port	Service	Version
22	ssh	OpenSSH 7.6p1 Ubuntu 4ubuntu0.5 (Ubuntu Linux; protocol 2.0)
30303	tcpwrapped	-

34.82.4.34

2.7 Standard Proxies found

Port	Service	Version
30303	tcpwrapped	-

138.197.145.243

Port	Service	Version
30303	tcpwrapped	-

134.122.38.219

Port	Service	Version
30303	tcpwrapped	-

104.196.67.0

Port	Service	Version
30303	tcpwrapped	-

34.65.48.0

Port	Service	Version
30303	tcpwrapped	-

138.197.134.217

Port	Service	Version
30303	tcpwrapped	-

206.189.8.187

Port	Service	Version

30303	tcpwrapped	-

134.122.38.223

Port	Service	Version
30303	tcpwrapped	-

62.133.229.44

Port	Service	Version
30303	tcpwrapped	-

34.82.248.206

Port	Service	Version
30303	tcpwrapped	-

143.198.32.242

Port	Service	Version
30303	tcpwrapped	-

165.22.195.253

Port	Service	Version
30303	tcpwrapped	-

3.131.130.187

Port	Service	Version
30303	tcpwrapped	-

34.65.249.131

Port	Service	Version
30303	tcpwrapped	-

143.198.38.130

Port	Service	Version
30303	tcpwrapped	-

95.216.100.80

Port	Service	Version
30303	tcpwrapped	-

143.198.46.146

Port	Service	Version
30303	tcpwrapped	-

35.236.12.156

Port	Service	Version
30303	tcpwrapped	-

34.89.70.153

Port	Service	Version
30303	tcpwrapped	-

35.185.80.25

Port	Service	Version
30303	tcpwrapped	-

143.198.33.127

Port	Service	Version
30303	tcpwrapped	-

159.203.56.216

Port	Service	Version
30303	tcpwrapped	-

143.198.33.123

Port	Service	Version

30303	tcpwrapped	-

138.68.28.193

Port	Service	Version
30303	tcpwrapped	-

138.197.143.145

Port	Service	Version
30303	tcpwrapped	-

5.255.64.12

Port	Service	Version
30303	tcpwrapped	-

62.133.229.47

Port	Service	Version
30303	tcpwrapped	-

34.89.176.47

Port	Service	Version
30303	tcpwrapped	-

134.122.37.139

Port	Service	Version
30303	tcpwrapped	-

34.94.100.243

Port	Service	Version
30303	tcpwrapped	-

195.201.129.23

Port	Service	Version
30303	tcpwrapped	-

143.198.38.120

Port	Service	Version
30303	tcpwrapped	-

138.68.22.108

Port	Service	Version
30303	tcpwrapped	-

34.205.255.255

Port	Service	Version
30303	tcpwrapped	-

80.71.57.43

Port	Service	Version
30303	tcpwrapped	-

35.195.170.62

Port	Service	Version
30303	tcpwrapped	-

34.121.81.57

Port	Service	Version
30303	tcpwrapped	-

34.82.59.126

Port	Service	Version
30303	tcpwrapped	-

142.93.159.37

Port	Service	Version

30303	tcpwrapped	-

34.106.150.151

Port	Service	Version
30303	tcpwrapped	-

143.198.32.44

Port	Service	Version
30303	tcpwrapped	-

138.197.153.21

Port	Service	Version
30303	tcpwrapped	-

85.17.90.39

Port	Service	Version
30303	tcpwrapped	-

54.179.195.18

Port	Service	Version
30303	tcpwrapped	-

165.22.225.173

Port	Service	Version
30303	tcpwrapped	-

34.83.169.70

Port	Service	Version
30303	tcpwrapped	-

165.22.195.242

Port	Service	Version
30303	tcpwrapped	-

34.125.86.16

Port	Service	Version
30303	tcpwrapped	-

165.22.203.135

Port	Service	Version
30303	tcpwrapped	-

34.236.52.203

Port	Service	Version
30303	tcpwrapped	-

63.32.69.102

Port	Service	Version
30303	tcpwrapped	-

34.68.235.86

Port	Service	Version
30303	tcpwrapped	-

104.198.91.187

Port	Service	Version
30303	tcpwrapped	-

3.135.113.157

Port	Service	Version
30303	tcpwrapped	-

167.99.215.162

Port	Service	Version

30303	tcpwrapped	-

143.198.46.115

Port	Service	Version
30303	tcpwrapped	-

34.85.140.179

Port	Service	Version
30303	tcpwrapped	-

46.137.133.42

Port	Service	Version
30303	tcpwrapped	-

64.225.118.195

Port	Service	Version
30303	tcpwrapped	-

35.233.134.218

Port	Service	Version
30303	tcpwrapped	-

143.198.40.23

Port	Service	Version
30303	tcpwrapped	-

94.130.108.20

APPENDIX A

POSSIBLE IP ADDRESSES FOR THE VALIDATOR CASE STUDY

- 34.105.80.234
- 34.105.88.61
- 34.105.117.188
- 34.105.124.115
- 34.127.7.193
- 34.145.116.148
- 35.185.193.40
- 35.197.44.53
- 35.197.62.120
- 35.197.69.100
- 35.197.108.92
- 35.199.185.185
- 35.212.129.63
- 35.212.234.25
- 35.212.254.197
- 35.230.33.100
- 35.230.33.136
- 35.230.60.49
- 35.230.92.52
- 35.230.116.151
- 35.233.152.68
- 35.247.36.111
- 35.247.58.13
- 35.247.78.68
- 35.247.120.79

THANK YOU FOR CHOOSING

// HALBORN